Roll No. : ....

MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: **B.Tech** Semester: VI

Name of the Course: Bigdata Storage and Processing

Course Code: TCS 671

Time : 90 Minutes **Maximum Marks : 50**

Note:

i. Answer all the questions by choosing any one of the sub questions
ii. Each questions carries 10 marks

Q1	(10 Marks)	CO1
(a)	What are the key technologies used in managing Big Data? How do they help in handling large-scale data efficiently? Explain.	
	OR	
(b)	Discuss the methods used for capturing Big Data. What are the challenges involved in this process?	
Q2	(10 Marks)	CO2
(a)	Explain the Big Data Architecture with help of suitable diagram.	
	OR	
(b)	Explain the big data characteristics with the help of six V's. How is digital data classified for bigdata processing? Explain in detail.	
Q3	(10 Marks)	
(a)	Discuss in detail the architecture of HDFS and its components. Explain how they work together to manage large-scale data storage.	CO1, CO2
	OR	
(b)	Compare HDFS with traditional distributed file systems. Highlight the key differences.	
Q4	(10 Marks)	CO2
(a)	Explain the Hadoop ecosystem with the help of a suitable diagram? Differentiate between RDBMS and Hadoop.	
	OR	
(b)	Differentiate the process of client reading data from HDFS and client writing data to HDFS with suitable diagrams.	
Q5	(10 Marks)	CO1
(a)	What is the role of NameNode and DataNode in HDFS? What is the purpose of the Command Line Interface in HDFS? Explain	
	OR	
(b)	Discuss the limitations of Hadoop in handling large-scale data processing in detail.	CO2